The Mental Model of an LLM – Part 1

Foundations of Large Language Model Understanding

WHAT WE'LL COVER TODAY

1

WHAT AN LLM ACTUALLY IS

DEFINING THE TECH & APPLICATIONS

2

HOW AN LLM THINKS

TOKEN GENERATION & CONTEXT

3

KEY CONCEPTS

ATTENTION, BIAS & PREDICTION

WHAT IS AN LLM ACTUALLY?

1. **NOT MAGIC**
2. **NOT A DATABASE**
3. **A PROBABILITY MACHINE**

TRAINED ON A FIXED SNAPSHOT

TRAINING COMMENCES

SEP 2023

TRAINING COMPLETION

1

TRAINED ONCE

THE MODEL PROCESSES DATA AS A ONE-TIME EVENT

2

KNOWLEDGE CUTOFF

ACCESS TO EVENTS ENDS AT THE SNAPSHOT DATE

3

NO REAL-TIME UPDATE

THE MODEL IS STATIONARY, NOT DYNAMIC (LIKE A HISTORY BOOK)

PROBABILISTIC GENERATION

1

NOT DETERMINISTIC

2

SAME PROMPT CAN GIVE DIFFERENT OUTPUTS

THE MODEL PREDICTS THE *NEXT* TOKEN STATISTICALLY

3

STATISTICALLY DRIVEN RESULTS

BASED ON TRAINED PATTERNS, NOT A FIXED LOOKUP

'FEATURE, NOT A BUG'

ENABLES VARIETY AND CREATIVE GENERATION
(vs. simple retrieval)

Why Same Prompt → Different Outputs?

1

THE 'TEMPERATURE' SETTING
CONTROLS THE DEGREE OF RANDOMNESS IN PREDICTION

2

LOW TEMPERATURE (<0.5)
FAVORS THE MOST PROBABLE NEXT TOKEN (DETERMINISTIC)

3

HIGH TEMPERATURE (>1.0)
ALLOWS LESS PROBABLE TOKENS (CREATIVE/DIVERSE)

VISUALIZING OUTPUTS ON THE TEMPERATURE SCALE

PROMPT:
A futuristic city in the desert...

Low (0.1) Predictable
A series of sleek towers with solar panels. **Focused and efficient design.**

Medium (0.7) Balanced
Bustling metropolises with hover-cars and sand-resistant domes. **Creative and functional.**

High (1.2) Diverse
Crystalline bio-habitats that glow with energy, powered by spice wind turbines. **Surreal and unexpected.**

How LLMs Read Text: Tokenization

1

BYTE PAIR ENCODING (BPE)
MERGES COMMON CHARACTERS INTO TOKENS
REDUCES VOCABULARY SIZE

2

SIMPLE WORD EXAMPLE
hello (1 TOKEN)

3

LONG WORD EXAMPLE
unprecedentedly
(MULTIPLE TOKENS)

VISUALIZING TOKENS: LEGO ANALOGY

Simple Word:
hello

Long Word:
unprecedentedly

un pre ced en ted ly

COMMON BLOCKS FORM RARE WORDS
BPE CREATES EFFICIENCY BY BREAKING DOWN UNFAMILIAR STRINGS INTO REUSABLE PIECES

Tokenization – Real Examples

1. Short Words

the cat sat

[the] [cat] [sat]

3 Tokens

2. Long/Rare Words

unpredictable

un pre dict able

4 Tokens

3. Code Snippets

```
def my_func(x):
```

[def] [my_] [func]

[(] [x] [)] [:]

7 Tokens

4. Emojis/Symbols

😄 Hello! 🔥

😄 [Hello] [!] 🔥

4 Tokens

Different LLMs tokenize text differently; examples based on common BPE models.

Context Window Explained

User Input: How does BPE efficiency relate to large words like 'unprecedentedly'?

LLM Response: BPE efficiency breaks 'unprecedentedly' into common sub-tokens (un-, pre-, ced, en, ted, ly), making processing rare words like...
(RESPONSE TRUNCATED - Context Limit Reached)

Everything must fit inside this window

Your Control Knobs

1. Temperature (Randomness)

- Controls randomness of output.
- Low = focused & predictable.
- High = creative & chaotic.

2. Top-p (Nucleus)

- Limits pool to tokens whose cumulative probability exceeds 'p'.
- Low p = focused, high-probability words.
- High p = more diverse options.

3. Top-k (Diversity)

- Selects only from the 'k' most likely tokens.
- Small k = precise, top choices.
- Large k = include less likely (but diverse) options.

Adjusting these parameters shapes the model's creative behavior.

Temperature – The Creativity Knob

1. Focused (0.0)

- Selects only high-probability tokens.
- Highly predictable, factual, and repetitive.
- Minimizes risk of nonsensical outputs.

Example Output:
Prompt: Continue: 'The old, rusty key turned slowly in the lock and...'
Output: ...the door opened to a dusty room with an empty desk and a

2. Balanced (0.7)

- Explores a broader distribution.
- Coherent and varied sentences.
- Maintains narrative logic.

Example Output:
Output: ...the door groaned open, revealing a hidden compartment and a forgotten diary, promising answers at last.

3. Creative (1.2+)

- Allows low-probability, surprising tokens.
- Unpredictable and innovative outputs.
- Higher chance of poetic and evocative language.

Example Output:
Output: ...suddenly the entire house dissolved into a shimmering vortex of memory and neon butterflies, tasting vaguely of electricity and regret.

Transformer at 30,000 Feet

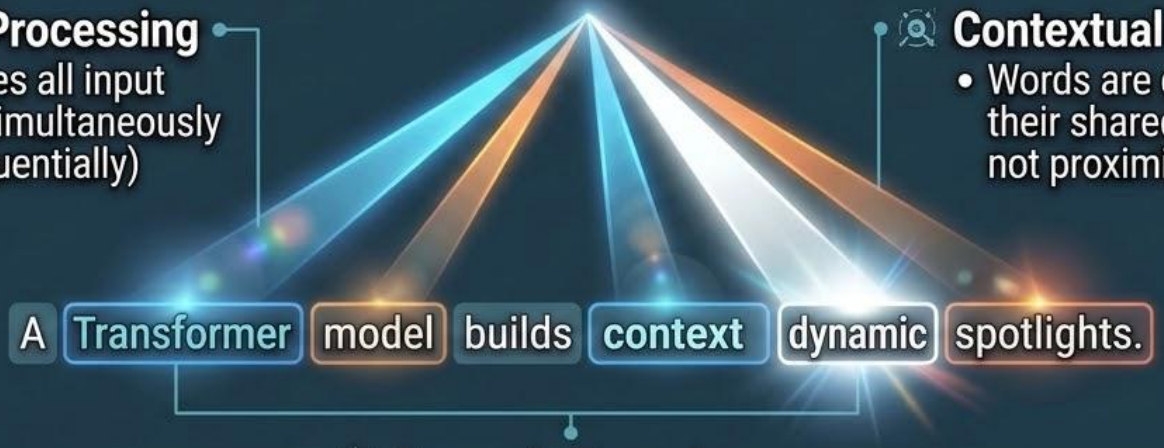
The Attention Mechanism

Parallel Processing

- Processes all input tokens simultaneously (not sequentially)

Contextual Relevance

- Words are connected by their shared meaning, not proximity



Dynamic Focusing

- Attention weight acts like a spotlight of varying brightness

*Focusing on core relationships for intuitive understanding.

The Lost in the Middle Problem

Input Context (100k+ Tokens)

Strong Primacy Effect
(High Attention)

Critical information here
is often neglected

Strong Recency Effect
(High Attention)

START

Critical information here is often neglected.

END

Phenomenon where models struggle to retrieve information from the middle of very long inputs.

⚠ **REAL-WORLD WARNING & PRACTICAL IMPLICATION:** Placement matters. Structure your prompts to put essential context or key instructions at the *beginning* or *end* of the input for maximum recall.

Key Takeaways – Part 1

- **Transformer Parallelism:** Inputs (like tokens) are processed simultaneously, not sequentially.
- **Dynamic Attention:** The core mechanism acts like a varying-brightness spotlight, building contextual relevance rather than proximity connections.
- **The 'Middle' Challenge:** Models struggle with information retrieval from the center of very long (e.g., 100k+ token) inputs.
- **Attention Inequity:** Recall is strongest at the start (Primacy) and end (Recency) of contexts, neglecting the middle.
- **Practical Prompting:** Position essential context and instructions at the beginning or end of prompts for maximum recall.

Questions?

Tomorrow: Reasoning Models & Model Comparison